Project Development Phase

Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Platform

Project Title	Script-Controlled ACL - Restrict Record Access Based on Field Value
Platform	ServiceNow (Table, Roles, ACL)
Team Size	5 Members
Lab Duration	3 hrs
Phase	5. Project Development Phase

Team Members

Member	Name	Role
Member 1	Swetha V	Team Lead
Member 2	Vaitheeshwari S	Requirements
Member 3	Swetha T	Table, Fields, Data
Member 4	Abinesh V	Users, Roles, ACL Scripts
Member 5	Anbuselvam M	Test Cases

1. Overview

This phase documents the step-by-step implementation carried out on the ServiceNow instance, following milestones M1 to M5.

Milestone 1 - User Creation

- Log in to the ServiceNow instance with admin access.
- Navigate to **User Administration > Users** and click **New**.
- User ID: EEEUser | First name: EEE | Last name: User | Email: eeeuser@gmail.com.
- Save the record.

Milestone 1 - Role Creation

- Navigate to **User Administration > Roles** and click **New**.
- Name: **bb1**, add a short description and click **Submit**.
- Repeat the same procedure for **bb2, bb3, bb4**.
- Assign all four roles to the EEE User.

Milestone 2 - Table Creation

- Go to **System Definition > Tables** and click **New**.
- Label: **Institution Details** | Name: **u_institution_details** | Extends: false. Click **Submit**.
- Add the fields below and save the table.

Field	Type

Student Roll Number	Auto Number
Student Name	Reference - User
Faculty Name	Reference - User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi-line String

Milestone 3 - Sample Data

Create multiple records in the Institution Details table with different Branch values (ECE, EEE, CSE). Example: (STU100, EEE), (STU200, ECE), (STU300, CSE).

Milestone 4 - READ ACL

- Elevate the **security_admin** role (profile menu > Elevate role).
- Go to **System Security > Access Control (ACL)** and click **New**.
- Type: record | Operation: read | Name: u_institution_details | Active: true | **Advanced: true**.

- In **Requires role** add **bb1**.
- Data condition: **Branch is EEE**.
- Paste the script below in the Script field and click **Submit**.

```
(function () {   // Allow admin users full access
if (gs.hasRole('admin')) {     return true;   }   //
Allow only users with role bb1 (EEE branch users)
if (gs.hasRole('bb1')) {     return true;   }   //
Deny access for all others   return false; })();
```

Milestone 5 - CREATE, WRITE and DELETE ACLs

ACL	Operation	Name	Requires Role	Data Condition
Create ACL	create	u_institution_details	bb2	None
Write ACL	write	u_institution_details	bb3	None
Delete ACL	delete	u_institution_details	bb4	None

For each ACL: click **New**, set Type = record, choose the operation, set Name = u_institution_details, keep Active = true, add the required role in the Requires role related list, and save.

Development Notes

- security_admin elevation ends when the user logs out; elevate again if required.
- The READ ACL script returns true for admin and bb1 users; the Branch = EEE filter comes from the data condition.
- All ACLs use the *Allow If* decision type with Admin overrides enabled.